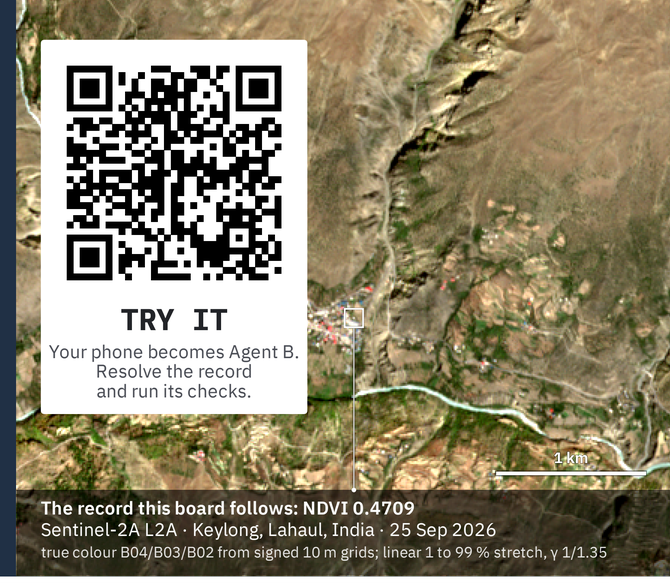
EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time.

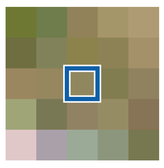
Jaya Kumari · **Avijeet Singh** · Vortx AI · avijeet@vortx.ai
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · ememdev · github.com/Vortx-AI/ememem (Apache-2.0)



1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.



Agent A reads NDVI 0.4709 and cites it
Sentinel-2A L2A, 25 Sep 2026
outlined: the 10 m pixel the record names
RGB from recorded grids

Agent A cites	handoff	relay	Agent B	B acted on corrupted evidence (agents, pooled Claude)
A	**prose** `The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Lahaul…`	relay or faulty signer changes: value, cell, time, band, source, derivation, stale state, signature, pixel	B reads the text	254 / 276
A	**JSON** `{"about": {"field": "Keylong, Lahaul (32.57126 N, 77.03448 E…`		B reads the fields	164 / 276
A	**retrieved text (RAG)** `Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell de…`		B retrieves a passage	220 / 276
A	**opaque id** `record ref obs-7d1c44`		B fetches the sender's record	154 / 288
A	**EMEM reference** `ememfact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67c…`		B resolve · re-hash · place / time / band · signature · log · recompute · source re-read	0 / 300

no check reaches
- the entity meant (M17)
- sensor accuracy
- the decision

GENUINE CONTROLS
71 / 72
expected decisions
checked reference · pooled Claude

In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

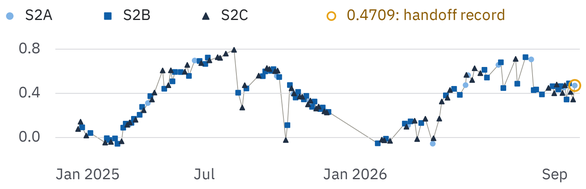
Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 as B, pooled; Qwen2.5-7B apart; 2,794 scored trials on 1 Oct 2026. Pooled results above are the Claude runs; Qwen is reported separately.
Qwen2.5-7B: prose 22 of 23; checked reference 2 of 25. One run. Simulated corruptions, constructed decision threshold; model breakdown and confidence intervals in Reproduce.

2 Can a time series retain its evidence?

NDVI history, with citations

Each value retains its dated record.

KEYLONG · SENTINEL-2 L2A · 141 RECORDS



Cloud/snow screening and processing harmonisation were not applied; physical interpretation needs both.

3 Where did the evidence change?

A changed value needs a cause

A cited value can change during a handoff. Between acquisitions, the surface, sensor or processing may also change.

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \varepsilon$$

Surface change, instrument, registration, encoder and residual. Δz is the later value minus the earlier one. The attribution ledger links evidence to each term; the numeric split remains open.

IMPLEMENTED · DRIFT-ANCHOR SCORE

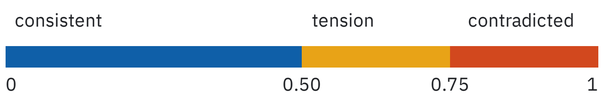
$$d = |x - a|$$

$$r = d / (3\sigma);\ s = r / (1 + r)$$

for finite x, a and $0 < \sigma < \infty$

Invalid σ: s = 0 if d = 0; otherwise s = 1.

x: output · a: anchor · σ: anchor uncertainty



Tests pin 3σ → 0.50 and 9σ → 0.75.

Score boundaries: consistent < 0.50; tension < 0.75.

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Use it in an EO workflow

Compare dates · recall, diff
Assemble evidence · merge, trace
Retain conflicts · competing, evolve
Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting product estimates and pass the cited evidence onward. Single-hop retrieval is implemented; multi-hop planning remains open.

20 upstream offline SDK / encoding tests passed.

Model outputs are records too

Satellite input → encoder → vector record → content address

Prithvi 1,024 values

checkpoint digest inside the record

TESSERA 128 values

product year 2024; no checkpoint digest

Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

THE CONTRIBUTION

Checkable observation handoff

An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.

Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

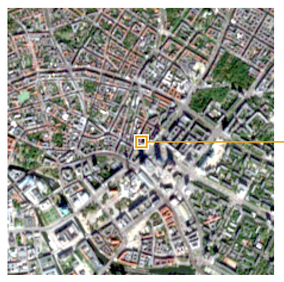
4 What does one Berlin location reveal?

One cell, many source records

One lookup joins the evidence; each product keeps its own time, provenance and native grid.

INSPECT Open the Keylong record and its 1,115 bytes.

BERLIN



`defi.zb655.yaka.pUxe`
Sentinel-2C L2A · 27 Sep 2026
RGB · 2.24 km across; box: 90 m
Marker locates the shared lookup cell.

direct sensor · deterministic index · model output · human curated · unclassified

product	attested reading	time / absent at	native grid	fact_cid
Sentinel-2C L2A	NIR 0.0475	27 Sep 2026	10 m	wtqd64…
Sentinel-2C L2A	NDVI 0.195	27 Sep 2026	10 m	y2k2ti…
Sentinel-1C RTC	VV −2.82 dB	28 Sep 2026	n/s	2x2enx…
GLO-30 DEM	36.6 m DSM	2021	30 m	sdu3gf…
MODIS MOD11A2	297.4 K day LST	29 Aug 2026	1 km	yqhihm…
ESA WorldCover	built-up (50)	2021	10 m	xicg6u…
ESA CCI Biomass	0 t/ha	2022	100 m	keebyc…
JRC GSW v1.4	0 % water	1984 to 2021	30 m	hohgux…
Hansen v1.13	0 % tree cover	2000	n/s	x4kpky…
JRC GFC2020 V4	not forest	2020	10 m	oog2b4…
Copernicus CAMS	NO_2 12.3 µg/m³	30 Sep 2026	~11 km	jvwhwg…
Overture, not EO	0 buildings	Aug 2026	vector	czehax…
ISRIC SoilGrids	*no data, null*	30 Sep 2026	250 m	mkwinv…
CHIRPS v2.0	*outside ±50°*	30 Sep 2026	0.05°	bkaovf…
JRC TMF	*outside ±30°*	30 Sep 2026	30 m	w7t2hx…
NASA FIRMS	*no fire in 24 h*	30 Sep 2026	points	k2ykb2…

15 products · 16 attested records · 4 signed absences · read 30 Sep 2026
Shared address, distinct native grids. Hatched: absence. n/s: grid not stated.
All 16 records re-hashed to their CIDs on 2 Oct 2026.

An EMEM cell indexes a location. Products retain their native pixels; CHIRPS records an out-of-coverage absence.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read

agents as B: three Claude models pooled, 1 Oct 2026
B acted on corrupted evidence · unaffected · refused · not applicable
M2 seen in production · the only check that stops it · ◆ flips the decision

			prose	JSON	RAG	opaque id	EMEM, full check (D E F G H I)	first check that refuses
	G0	nothing altered: accepted everywhere, never refused						
value	M1	stated value moved by 1 ULP	12/12	12/12	12/12	9/12	C L0	
	M2	stated value rounded to "0.47"	6/12	6/12	11/12	0/12	C L0	◆
	M8	value 0.45, re-encoded and re-hashed	12/12	6/12	12/12	6/12	F L0	◆
cell	M4	record cited for another cell	7/12	0/12	6/12	6/12	E L1	
	M9	cell changed inside the record, re-hashed	7/12	11/12	6/12	6/12	F L0	
time	M10	tslot changed to look current, re-hashed	12/12	6/12	6/12	6/12	F L0	
band	M6	a record for another band handed over	12/12	0/12	6/12	2/12	E L1	
source	M11	source scene id changed, re-hashed	12/12	12/12	12/12	12/12	F L0	
derivation	M12	offset changed, value recomputed, re-hashed	12/12	11/12	12/12	8/12	F L0	◆
	M14	value (0.46) disagrees with the signed DNs	12/12	6/12	12/12	6/12	H L2	◆
stale / current	M5	an older record handed over as current	12/12	1/12	6/12	1/12	E L1	
	M16	a second signed version shown only to B	12/12	6/12	6/12	6/12	G L0	
signature / id	M3	1 ULP changed inside the served bytes	12/12	12/12	12/12	12/12	D L0	
	M7	token miscopied by one character	n/a	n/a	n/a	0/12	C L0	
	M13	value 0.45 under the forger's own key	12/12	6/12	12/12	6/12	F L0	◆
source pixel	M15	DNs read from the pixel 10 m south	12/12	12/12	12/12	12/12	I L3	◆
entity	M17	same record; A meant another entity (out of scope)	not run	not run	not run	not run	none	

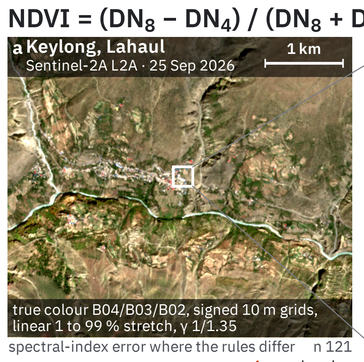
B acts on corrupted evidence
254 / 276 (R1 15/15) · 164 / 276 (R1 15/15) · 220 / 276 (R1 13/16) · 154 / 288 · 0 / 300 (R1 0/16)
C resolve L0 · D hash L0 · E binding L1 · F signature L0 · G log L0 · H recompute L2 · I re-read L3 · D to I are cumulative: each adds one check
Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026.
Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 A valid record can preserve a wrong source read

$$NDVI = (DN_8 - DN_4) / (DN_8 + DN_4 + 2o),\ o = -1000 \text{ (product offset)}$$



spectral-index error where the rules differ n 121
0.0001 0.001 0.01 0.1
median 0.027 · p90 0.113 · max 0.301

b NDVI of each 10 m pixel

0.59	0.65	0.49	0.41	0.37
0.53	0.55	0.50	0.50	0.39
0.39	0.43	0.47	0.54	0.44
0.30	0.33	0.30	0.41	0.48
0.09	0.06	0.07	0.21	0.43

c test rule: irrigate if NDVI ≤ 0.4705
0.4709 → hold named pixel
0.3016 → irrigate pixel 10 m south

hash · binding · signature · log · recompute · re-read
pass the wrong value · refuses

162 of 200
sampled pre-fix records carry a neighbour's values
Wilson 95 %: 75 to 86 %
38: both rules read one pixel

the real record `kxjvfwpa…`: 23 Sep 2026 · signed **0.3444** · its containing pixel **0.4860**
Sample: 200 pre-fix Sentinel-2 records cited in ememdev's public channel, one per cell, seeded; 192 Element84, 8 Planetary Computer. After the fix, 0 of 54 (Planetary Computer, two days). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.

7 What has the receiver established?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Checks establish properties of the cited record. A source re-read adds evidence; sensor accuracy remains inherited.

8 Can a later agent recover the earlier state?

Revisit the earlier evidence

$C_\tau = TxAsOf(O^*, a, b, \tau)$

$recall = \max_{(t, cid)} \{O \text{ in } C_\tau : t \le t^*\}$

Latest-as-of mode: take the versions known by signing time τ, then the latest valid time ≤ t*. No candidate gives an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

same place + band

signed 28 May	signed 11 Aug
918.0 m	**915.07 m**
Open-Meteo / DEM90	Copernicus DEM30
earlier CID	later CID

as of 15 Jun → 918.0 m

The next agent can recover the earlier citation.

As of 15 Jun → 918.0 m. The later 915.07 m is a provider change. Temporal storage is established; the handoff adds a portable citation to the exact earlier record.

9 What does the memory contain?

Observations + temporal edges

$M = (O^*, E^*)$ $O = (a, b, t, v, u, p, s)$

a: location cell · b: variable · t: valid time · v: value
u: uncertainty · p: provenance, recipe, signing time
s: attestation associated with the observation

Conceptual tuple: a batch attestation covers the fact address; a signed read receipt binds the response.

$E = (subject, predicate, object, t_{from}, t_{to})$

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t)
$cid(record) = BLAKE3(CBOR_{canonical}(record))$

10 What else can an agent carry?

Beyond a single observation

data need	grammar, after emem:	what its id hashes
a place	`cell:<cell64>`	the quantised lat/lng; no hash
one value	`fact:<cell>:<cid>`	BLAKE3(CBOR(fact)), 52 chars
nothing there	`fact: · kind absence`	same; + BLAKE3(reason)[:16]
an embedding	`fact: · 128 / 1024 floats`	same; attester_only
a derived value	`fact: · op, parents`	same; pure ops re-runnable
a raster	`raster:<a>:<b>:<t>:<d>`	fact d names BLAKE3(grid)
a raster in time	`cube:<a>:<b>:<t0..t1>:<d>`	BLAKE3(CBOR([member d…]))
a raster set	`rasterset:<set>:<d>`	BLAKE3(CBOR([d…, purpose]))
many facts, one id	`bundle:<cid>`	BLAKE3(p, cell\|band\|t\|cid)[:16]
a file, one chunk	`tree:<file>#row=i`	BLAKE3(note)[:16]; root inside
a reasoning stage	`state:<cid>`	BLAKE3(CBOR(state record))
a named object	`entity:<cid>`	BLAKE3(anchor)[:16]: a label
a device run	`trace:<cid>`	BLAKE3(CBOR(signed trace))
an evidence chain	`track.v1 note`	link = BLAKE3(prev, ref)[:16]

14 data needs; embedding and absence share the fact grammar.
Hash rules: emem 18adb67. A name, a set and bytes differ.
track.v1 is a note. Trace evidence here is a reference harness.

Eight facts resolve through one bundle; four checkpoint hashes reproduce. These object checks do not validate an agent's reasoning.

NEXT EXPERIMENTS

Measure each cause of change

Control scene, pixel and encoder changes to test numeric attribution. Connect device outputs to separately recalled anchors. Repeat handoffs across separately operated hosts.

11 Use emem. Carry the evidence forward.

Start with an EO question, connect your tool, then pass the cited observations onward. Public emem reads need no API key.

Measured separately: the same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).
The routes below are plugins, connectors, packages and listings. One evidence protocol, multiple agent runtimes. Setup and compatibility notes are behind CONNECT.

CONNECT Choose an EO task, copy a reference, run the checks. `vortx-ai.github.io/esa_poster/use/`

Attested observation → **MCP / A2A** → **Agent host / client** → **Next agent / application**

ChatGPT
@emem plugin
Enable emem; paste a token.

Claude
Code plugin / MCP
Resolve from chat or terminal.

Visual Studio Code
MCP connector
Copilot agent mode: @mcp emem

Dify
Marketplace plugin
Referent Lock / EUDR workflows

Salesforce MuleSoft
Exchange listing
Open Vortx AI MCP Server

MORE CLIENTS
Cursor
Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript
Docker · LangGraph memory
GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex
AutoGen · CrewAI · Mastra
Agno (see setup notes)

DISCOVERY / MIRRORS
GitHub MCP · MCP Registry
Glama · Smithery · ClawHub
Context7 · Hugging Face (older mirror)

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)